INCH-POUND

MIL-W-80211C
24 October 1990
SUPERSEDING
MIL-W-80211B
19 September 1985

MILITARY SPECIFICATION

WELDING MACHINE, (MICROWELDER), RESISTANCE TYPE, STORED ENERGY, CONSTANT VOLTAGE, LOW IMPEDANCE SOURCE, BENCH MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and one size stored energy, resistance welding machine consisting of a power supply and various configurations of welding heads.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-8939 - Welding, Resistance, Electronic Circuit Modules.

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARD

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

(Unless otherwise indicated copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards. (DOD adopted)

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103-1403.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete welding machine shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Design. The welding machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The welding machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the welding machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical and pneumatic operating controls shall be located convenient to the operator's work station.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the welding machine, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910. Additional safety and health requirements shall be as specified in detail (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial welding machine supplied to the commercial

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market shall be provided for the point of operation and all other parts of the welding machine that present safety hazards.

3.2.6 Mercury restriction. The welding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the welding machine, and its components materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the welding machine to its maximum rating and capacity without permanent distortion, or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the welding machine shall be painted in accordance with the manufacturer's commercial practice and color.

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3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4 Components. The welding machine shall consist of not less than an enclosure, a power supply, an electrical system, a welding head, and electrodes.

3.4.1 Enclosure. The enclosure shall be constructed of corrosion resistant metal or metal treated to resist corrosion. The power supply and electrical system shall be housed in the enclosure. When specified (see 6.2), ventilation shall be provided by a 120 volt, 60 Hz, single phase forced air blower. When necessary the air inlet shall be provided with a replaceable or cleanable filter. The enclosure shall have provisions for inspection and maintenance.

3.4.2 Power supply. The power supply shall include a low impedance current source, a charging circuit, and a voltage regulator. The power supply shall be of the constant voltage type, and shall be capable of producing consistently reproducible welds at any given control setting within the maximum peak ampere capability of the power supply.

3.4.2.1 Low impedance source. The low impedance source shall be either a sealed, dry, lead-acid battery designed to furnish a no load voltage of 7.35 volts, or a capacitor bank. The source shall be designed to furnish not less than 500 amperes for 18 milliseconds.

3.4.2.2 Charging circuit. The charging circuit shall automatically begin to charge the power supply after approximately 4 welds have been made and shall maintain the power supply in a fully charged state.

3.4.2.3 Voltage regulation. The power supply shall maintain the voltage within ± 3 percent throughout the welding range. The regulation response to line or load variations shall be within 500 milliseconds.

3.4.3 Electrical system. Unless otherwise specified (see 6.2), the electrical system shall conform to NFPA 70. Each welding machine shall draw all of its electrical power from a single 120 volt single phase, 60 Hz circuit.

3.4.3.1 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.3.2 Controls. The controls shall have the functions necessary to assure that the machine can produce welds in accordance with Military Specification MIL-W- 8939. All calibration data shall be clearly marked in standard terms. When furnished, all meters shall have an accuracy of ± 2 percent of full scale reading.

3.4.3.2.1 Control circuit voltage. Auxillary control circuits shall be isolated from the input electric power supply by a low voltage transformer having a secondary voltage no greater than 24 volts.

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3.4.3.3 Bus bar and cable connections. Connections for bus bars and cables shall be of the plug-in or bolted type. Connections for weld cables shall be keyed, polarized, or use an indicator light to indicate voltage feedback polarity is out of phase, to ensure proper relationship.

3.4.3.4 Power supply cord. The power supply unit shall be furnished with not less than 5 feet of three-conductor type SO flexible electric cord in accordance with NFPA 70. The cord and plug shall be of the proper capacity, temperature, and voltage rating for the intended use. The cord shall have a strain relief connector where it enters the enclosure. The third wire shall be grounded to the frame of the unit.

3.4.3.5 Main disconnect switch. The welding machine shall have a main fusible disconnect switch, a circuit breaker or both that will serve as an "on-off" switch and as an overload protective device (see 6.2). A pilot light with red lens shall be furnished to indicate when the power is energized.

3.4.4 Welding head. The welding head shall be compatible with the power supply unit for performing various welding and bonding processes and shall be attached to the power supply by plug-in or bolted type connections. The welding head shall support the welding electrodes and their holders. The welding head ram movement shall have fast acceleration, low inertia, and low friction. The electrode force shall be calibratable and adjustable. Weld current shall flow only after the preselected force has been obtained. The repeat operation shall be within ± 3 percent of the setting. Electrode force shall be independent of operator variables. The welding cycle and action of the head shall be controlled by a foot pedal, unless otherwise specified (see 6.2), when the ram is air actuated a hand or knee type actuating control may be specified (see 6.2), the machine shall be furnished with a hand or knee type actuating control only when the ram is air actuated. Unless otherwise specified (see 6.2), the welding head shall be of the parallel gap type.

3.4.4.1 Electrodes. Unless otherwise specified, the welding machine shall be equipped with one set of parallel gap electrodes conforming to RWMA Bulletin Number 16, group A, class 2 alloy. When specified (see 6.2), electrodes of the quality, type, size, and class of alloy required shall be furnished.

3.4.4.2 Attachments and accessories. The welding head shall provide for attachments and accessories to be used to make up a complete system such as lighted optical aids, heat sensors, baseplates, X-Y positioners, or special accessories. The attachments and accessories shall be as specified and shall be fully described (see 6.2).

3.5 Performance. The welding machine shall be capable of precision microwelding small wires and leads on miniature electronic components in accordance with MIL-W-8939.

3.5.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.6 Standard equipment. All standard equipment normally provided with the manufacturer's commercial welding machine shall be furnished.

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3.7 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.8 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.9 Marking on instruments, control panels, charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.9.1 Nameplate. A nameplate shall be securely attached to each welding machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.10 Workmanship. Workmanship of the welding machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.1.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The welding machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each welding machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. Each welding machine shall be operated at no load for not less than 15 minutes. Proper operation of all controls, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. After the test in 4.7.1 has been completed and the welding machine is at its normal operating temperature, the welding machine shall be tested for welding capability in accordance with the requirements of MIL-W-8939. There shall be no evidence of malfunction of the welding machine.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirement for packaging shall be in accordance with MIL-W-45562 (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The equipment covered in this specification is intended for use in contractor's plants, military and government repair facilities for precision microwelding applications such as welding integrated packages to circuit boards, and small wire or ribbon to module interconnection. Weld applications for this equipment is found in thin film lead welding, and fine wire welding.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. First article, if required (see 3.1).

d. Specify system of units for measuring and indicating devices (see 3.2.1).

e. Additional safety and health requirements, if required (see 3.2.5).

f. Painting, if different (see 3.3.5.).

g Ventilation fan, if required (see 3.4.1).

h Electrical system and input power, if different (see 3.4.3).

i. Main disconnect switch, if different (see 3.4.3.5).

j. Foot actuating control, if required (see 3.4.4).

k. Hand or knee actuating control, if required (see 3.4.4).

l. Welding head, if different, specify type required (see 3.4.4).

m. Electrodes, if different, specify quantity, type, size, and class of alloy required (see 3.4.4.1).

n. Welding head attachments and accessories, if required, describe fully (see 3.4.4.2).

o. Additional performance features, if required (see 3.5.1).

p. Specify and fully describe optional equipment, if required (see 3.7).

q. Specify applicable repair parts, if required (see 3.8).

r. Nameplate, if different (see 3.9.1).

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S. Inspection conditions, if different (see 4.3).

T. First article inspection, if different (see 4.4).

U. Quality conformance inspection, if different (see 4.5).

V. If packaging, packing, and marking in accordance with MIL-W-45562 is required, specify level required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the remarks section of the DD Form 1423 that all technical data furnished should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 INCH-POUND units. INCH-POUND units are a system of measures based on the yard and pound commonly used in the United States of America and defined

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by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.8 Training. Training required by the procuring activity should be provided as specified in the contract.

6.9 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.10 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.11 Subject term (key word) listing.

Battery

Capacitor

Parallel gap

Recharge

Custodians:
Army - AL
Air Force - 99

Review activities:
Air Force - 84

User activities:
Navy - MC

Preparing activity:
DLA - IP

Project 3432-0141

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-80211C	2. DOCUMENT DATE (YYMMDD) 24 October 1990

3. DOCUMENT TITLE: Welding Machine, (Microwelder), Resistance Type, Stored Energy, Constant Voltage, Low Impedance Source, Bench Mounted

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
WILLIAM KENNON/DIPEC-SSG	(901) 775-6989	683-6989
c. ADDRESS (Include Zip Code) Defense Industrial Plant Equipment Center 2153 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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